

NOTE: Key messages to watch for: MSG 3 = X Driver Alarm | MSG 4 = Y Driver Alarm | MSG 8 = No Air Pressure | MSG 9 = Zero Machine Not Searched (homing incomplete).

TASK 3 CHECK PLC INPUT SIGNALS — I/O DIAGNOSTIC SCREEN

1. In Automatic mode, press the < or > arrow keys at the bottom of the screen to navigate to screen 4/5, then 5/5.
2. Green highlighted background on an input means it is reading 1 (active).
3. Record the actual value (0 or 1) in the table. Circle the value you see on screen.
4. Compare against the Expected column. Any mismatch is a clue.

Input	Name	Expected	Actual (circle)	Notes / Action if wrong
I01	EMERGENCY	1	0 1	Must be 1. If 0, E-stop circuit is open.
I02	SYSTEM READY POWER SUPPLY (1)	1	0 1	If 0, check main power supply contactor.
I03	AIR PRESSURE SWITCH	1	0 1	If 0, MSG 8 will also show. Check compressor / air line.
I04	OIL FILTER FILL IN	1	0 1	If 0, MSG 6 (Oil Filter Dirty) may also be present.
I05	AXIS X HOME SWITCH	—	0 1	During homing this must pulse to 1. Record value BEFORE and AFTER homing attempt.
I06	X AXIS DRIVE READY	1	0 1	If 0, X servo drive has lost power or fault.
I07	X AXIS DRIVE ALARM	0	0 1	Should be 0 (no alarm). If 1, X drive fault active. Try RESET.
I08	AXIS Z HOME SWITCH	—	0 1	Should pulse to 1 when Z reaches home during homing.
I09	C AXIS LOCK CYLINDER DETECTOR	0	0 1	C axis (shear rotation). Record as found.
I10	PUNCH Z BACKWARD DETECTOR	1	0 1	Z punch should be in back (retracted) position. If 0, punch is still extended.
I11	PUNCH Z FORWARD DETECTOR	0	0 1	Should be 0 when punch is retracted. If 1 simultaneously with I10, sensor fault.
I12	AXIS Y HOME SWITCH	—	0 1	Must pulse to 1 during Y axis homing.
I13	Y AXIS DRIVE READY	1	0 1	If 0, Y servo drive fault.
I14	Y AXIS DRIVE ALARM	0	0 1	Should be 0. If 1, Y drive fault active. Try RESET.
I18	CARRIAGE MATERIAL DETECTOR	—	0 1	0 = no material on carriage. Record as found.
I36	SECURITY PHOTOCELL 1	1	0 1	Safety light curtain. Must be 1 (clear) to operate.
I48	SECURITY PHOTOCELL 2	1	0 1	Safety light curtain. Must be 1 (clear) to operate.

NOTE: Your last observed values (from photos): I07=1 (X ALARM ACTIVE), I14=1 (Y ALARM ACTIVE), I05=0 (X not at home), I12=0 (Y not at home), I10=1 (Z punch retracted OK), I11=0 (OK). Priority: clear I07 and I14 before attempting homing.

TASK 4 ATTEMPT DRIVE ALARM RESET

I07 and I14 showed drive alarms active. These must be cleared before homing will succeed.

1. Press the RESET button on the Fagor 8070 keyboard (large button, lower right of keypad cluster).

2. Watch the status bar — does any MSG 3 or MSG 4 (X/Y Driver Alarm) disappear?
3. Go back to the I/O screen (Task 3) and re-check I07 and I14. Record new values.
4. If alarms DO NOT clear with RESET: power OFF the entire machine from the main electrical cabinet switch.
5. Wait 30 seconds, then power back ON. Allow the PC-CONTROL to boot fully.
6. Re-check I07 and I14 after restart.

Check	Before RESET	After RESET	After Power Cycle	Expected	Action if still wrong
I07 – X AXIS DRIVE ALARM				0	Note alarm code on X drive display. Contact GEKA.
I14 – Y AXIS DRIVE ALARM				0	Note alarm code on Y drive display. Contact GEKA.
MSG 3 in status bar				Gone	X drive hardware fault.
MSG 4 in status bar				Gone	Y drive hardware fault.

NOTE: The servo drive units have their own small LED display showing a fault code. If RESET and power cycle do not clear I07 or I14, photograph the code shown on the face of the X and Y servo drives inside the electrical cabinet before calling GEKA technical support.

TASK 5 ATTEMPT FULL MACHINE HOMING (CALIBRATION)

The manual calls this 'Calibrating the Machine'. It must be run every time the machine is switched on. MSG 9 (Zero Machine Not Searched) must clear after this.

1. Confirm drive alarms are cleared (I07 = 0, I14 = 0) from Task 4 before continuing.
2. Turn the Manual/Automatic key switch to AUTOMATIC position.
3. On the CNC keyboard, select MANUAL operating mode (hand icon — do NOT confuse with the key switch position).
4. Press the ZERO key on the keyboard.
5. Press the START button (green, lower right of main panel).
6. The machine will run a homing subroutine — all three axes (X, Y, Z) will move to their reference positions in sequence. Stand clear.
7. Watch the I/O screen: I05 (X HOME), I08 (Z HOME), and I12 (Y HOME) should each pulse to 1 as each axis passes its home switch.
8. After the routine completes, check whether MSG 9 (ZERO MACHINE NOT SEARCHED) has disappeared from the status bar.

Home Switch	Pulsed to 1 during homing?	MSG 9 cleared after?	Notes
I05 – X AXIS HOME SWITCH	YES / NO	YES / NO	
I08 – AXIS Z HOME SWITCH	YES / NO	YES / NO	
I12 – AXIS Y HOME SWITCH	YES / NO	YES / NO	

NOTE: If the homing routine stops mid-sequence or one axis does not reach its home switch, note WHICH axis stalled and any message that appears. This pinpoints whether the problem is the X carriage, Y punch head, or Z home sensor.

TASK 6 IF HOMING FAILS — IDENTIFY WHICH AXIS AND WHY

If MSG 9 remains after the homing attempt, one or more axes did not find their reference switch. Use the table below to isolate which axis and the likely cause.

Axis	Home switch input	Did it pulse?	Things to check physically
X (Carriage)	I05	YES / NO	Is the X carriage physically obstructed? Check the X home proximity sensor is mounted correctly and its LED lights when the carriage is at the home position. Check sensor cable.
Z (Horiz. Punch)	I08	YES / NO	Check Z axis home proximity sensor. I10 should be 1 (punch retracted). If I10=0, the punch cylinder has not returned fully — check hydraulic pressure and Z cylinder solenoid.
Y (Punch Head)	I12	YES / NO	Check Y axis home proximity sensor and encoder cable. I13 (Y Drive Ready) must be 1. If Y drive alarm (I14) is still on, Y cannot home.

NOTE: Proximity sensor LEDs: each sensor has a small indicator LED on its body. When the sensor detects its target (the axis is at home), the LED illuminates. You can jog manually (carefully, in small steps) toward the home position to see if the LED lights — if it does but the CNC still reads 0, the cable or PLC input card is the fault. If the LED does NOT light at all when physically positioned at home, the sensor itself is faulty.

TASK 7 VERIFY: RE-TEST Z– AND X+ AFTER SUCCESSFUL HOMING

Once MSG 9 has cleared (all axes homed), retry the original movements that triggered MSG 7.

1. Confirm the status bar shows NO active messages (or only non-critical ones).
2. In Manual mode, attempt to jog Z– (punch down) slowly in small steps.
3. In Manual mode, attempt to jog X+ (carriage forward) slowly.
4. Record results below.

Movement	MSG 7 still appears?	Machine moves?	Notes
Z– (punch down)	YES / NO	YES / NO	
X+ (carriage forward)	YES / NO	YES / NO	

NOTE: If MSG 7 clears and both movements work after successful homing: root cause was incomplete axis referencing. Investigate WHY homing was not completing previously (drive alarms on startup, home switch condition, startup procedure not followed). If MSG 7 still appears even after all axes homed: escalate to GEKA technical support — there may be a PLC parameter or Z-axis position threshold issue.

TASK 8 ESCALATION — INFORMATION TO GATHER BEFORE CALLING GEKA

If the fault cannot be resolved on site, collect the following before contacting GEKA Maquinaria technical support:

	Information to collect	Value / Notes
1	Machine serial number (plate on machine frame)	
2	CNC software version (visible on startup screen)	
3	Full list of active PLC messages (from Ctrl+M)	
4	Fault code shown on X axis servo drive display (inside electrical cabinet)	
5	Fault code shown on Y axis servo drive display (inside electrical cabinet)	
6	Which home switches pulsed and which did not during homing attempt	
7	I07 and I14 values before and after RESET and power cycle	

8	Any recent changes — new program, punch change, maintenance, power cut	
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